Applying Data Analytics Principles for Human Resource Analytics: Employee Turnover Prediction and Workforce Segmentation

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The studies focus on business related problems, such as factors affecting the working state of the workers and ultimately predicting employee attrition. In this work, Data Analytics study is applied in Human Resource Management (HRM) with the dataset which is the personnel record of a global company. The models employed were two machine learning classifiers; namely, Random Forest Classifier and Decision Tree Classifier for predicting employment status [17] and [22]. The overall accuracy of the Random Forest Classifier was 54% and feature significance was 95.8% for Salary_INR and F1-Score of the Active class was 0.70. The Decision Tree Classifier performed an overall accuracy of 69% with feature significance of Salary_INR as 61.0%, F1-Score as 0.81 for Active class. The findings offered some suggestions to the organisation to decrease employee turnover.

Keywords—*HR Analytics, Employee Attrition, Random Forest Classification, Decision Tree Classification, Feature Importance, Predictive Modeling.*

I. Introduction

Analytics for human resources (HR) is an important field in contemporary businesses. Converting a routine process into a data-driven, strategic decision-making engine [10], [11]. Due to recruiting costs and a loss of organizational momentum, large international firms often experience financial and operational losses as a result of staff turnover [14]. Conventional HR procedures, such examining departure interviews, are based on a more reactive structure. Predictive data analytics, on the other hand, enables firms to assess compensation distribution and detect turnover concerns, which should have a positive effect on organizational performance [9], [15]. HR analytics integration must, however, also be able to handle moral dilemmas pertaining to the administration of employee data [5].

Using a sizable HR dataset from the website Kaggle [1], which has 2,000,000 records with 12 different characteristics, this study examines employee data. Two main business questions are addressed by the analytical pipeline: (1) what factors have the greatest impact on operational employee status; and (2) does salary serve as a dependable and structured metric across employee status. This study identifies systemic trends in the distribution of salary and performance evaluation through the use of classification models, together with data visualization techniques to forecast employee attrition [6] and data mining to predict performance [2].

II. Related Work

The focus of predictive modeling in employee management has definitely moved towards machine learning frameworks. Comparative studies of the machine learning models have shown that the models with tree architecture, for example, Random Forest algorithms, outperform the traditional models in the case of the issues about an employee's status [17], [22]. Additionally, AI has created new avenues for strategic workforce planning in the HR management process [23]. Studies on sustainable HR shows that psychological well-being is one of the foundations in shaping the performance of an organization [7]. More recent challenges include work-life balance and burnout amongst the newer generation of workers further emphasize the need of achieving job satisfaction [3].

There are a number of recent studies on workplace transition during the pandemic that are showing a widening assessment gap between on-site and remote working. Hybrid working from home has proven to have a positive effect on retention without negatively impacting objective performance [4]. Pay is still a subject of debate and recent research into psychology has studied whether compensation is truly a motivator for well-being and performance or if there is a need to meet additional psychological needs [8]. To speed up these investigations, datasets for predictive modelling of attrition

have been made available to the research community [18, 25]. This research is an extension of the known methodologies with a predictive, exploratory pipeline for assessing structural alignment of a large enterprise system.

III. Dataset and Data Preparation

A. Dataset Specifications

We worked with a rather big dataset called "HR Data MNC" [1], which contains entries for 2,000,000 employees spread across 12 columns. Numerical data (such as Salary_INR, Experience_Years, and Performance_Rating), category information (such as Department, Job_Title, Location, Status, and Work_Mode), a date field (Hire_Date), and a few identifier columns are all captured by these columns. A couple things jump out when you look at the stats up front. With an average of around 896,888 INR and a standard deviation of about 402,610 INR, salaries were all over the place, indicating a wide range. The distribution of years of experience is quite uniform, ranging from 0 to 15. A well balanced distribution across the workforce is suggested by performance evaluations, which are based on a scale of 1 to 5 and do not significantly skew toward either extreme.

B. Data Preprocessing Pipeline

The raw data had to be processed before any analysis could be started. There are 5 steps to get it into shape.

1. Three columns were dropped right away: Unnamed: 0, Employee_ID, and Full_Name. None of them carry any analytical value, so keeping them around would just add noise.

2. Hire_Date was changed to datetime64 from a plain string. If it had been left unaltered, it would have posed issues when we came to feature engineering.

3. We then ran a quick data quality check across all 12 columns. No nulls came up, and there were no duplicate rows either, so no cleaning was needed on that front.

4. We used the IQR approach on the numerical columns to identify outliers, eliminating any values that did not fall within the range [Q1 − 1.5×IQR, Q3 + 1.5×IQR].

5. Next, we included two interaction variables (Experience × Performance and Tenure × Performance), a consistent encoding for the country/location field, and Tenure_Years derived from Hire_Date to give the model additional signal.

IV. Exploratory Data Analysis

A. Univariate Distributions

The IT department comes out on top with close to 500,000 employees, ahead of Sales and Operations. More interesting is the spread across the Status categories. Active employees make up roughly 1.2 million records, Resigned employees account for around 400,000, and the Retired and Terminated groups are considerably smaller. That kind of imbalance is worth flagging early since it could affect classification performance down the line. Work mode, by contrast, sits at about 58% on-site and 42% remote, which is a much more even split.

B. Bivariate Relationships

Several things in the bivariate analysis didn't quite add up. Performance ratings looked nearly identical across all departments, which is unusual and suggests the variable doesn't carry much real variation. A box plot of Salary_INR against Performance_Rating backed this up: medians and ranges overlapped heavily across all rating groups, so higher-rated employees weren't really earning more. Checking employee status against performance ratings showed the same thing, with Active, Resigned, Retired, and Terminated categories all turning up similar proportions at every rating level.

V. Methodology

A. Multi-Class Status Classification

The models used to predict the status of the employee (Active, Resigned, Retired, or Terminated) were Random Forest Classifier and Decision Tree, which were trained using an 80/20 split. To avoid the model being influenced by any one variable, categorical variables were encoded using LabelEncoder and numeric variables were scaled using StandardScaler, which were applied prior to the model being fed into.

VI. Results and Discussion

Two classification models were used for predicting the employee status as mentioned above, both trained on the same data set with the same preprocessing steps: Random Forest Classifier and Decision Tree model.

A. Results From RandomForestClassifier

The overall predictive accuracy obtained with the Random Forest Classifier was 54%. Currently the predictive capacity of the model is restricted due to the high class imbalance [16]. The model performed well for most of the 'Active' class (F1 score of 0.70) but poorly for the remaining classes, with the 'Resigned' class having an F1 score of 0.20 and the 'Retired' and 'Terminated' classes being 0.05. The confusion matrix also

shows that the classes that are regarded as the minority classes are misclassified with the majority class being ‘Active’. The feature importance analysis shows that ‘Salary_INR’ is the most important feature contributing 95.8% and ‘Experience_Years’ contributing 2.4%, and ‘Job_Title’ and ‘Performance_Rating’ contributing 0.7% each.

B. Results From Decision Tree Model

The overall accuracy was slightly better with the Decision Tree Classifier (69%), but this is only said at the overall level and is misleading at the class level. As displayed in the classification report, the model for the class "Active" achieved a precision of 0.70, a recall of 0.97 and an F1 score of 0.81, which means that the model predicts "Active" in almost all of the cases. The other two classes, however, were much harder to classify, with "Resigned" getting an F1-score of only 0.02 and "Retired" and "Terminated" both 0.01, indicating poor overall performance. Additionally, the confusion matrix confirms that the majority of all true labels for each class, regardless of the class, were misclassified as “Resigned” or as various other classes that were not the true class, with the diagonal (correct prediction) being surprisingly low for all classes other than “Active.” This apparent increase in accuracy is further confirmed by the macro average F1-score of 0.22, which is not a true measure of discriminative learning but rather due to class imbalance.

The feature importance of the Decision Tree showed a more scattered and yet imbalanced distribution. Feature f5 accounted for 60.8% of importance, followed by f3 at 13.2%, f1 at 9.1%, f2 at 7.6%, f0 at 5.2%, and f4 at 4.1%. The dominance of a single feature is similar to the Random Forest and indicates that one variable (probably Salary_INR) continues to split the data in an inexplicable way.

VII. Key Findings and Insights

A. Severe Class Imbalance

Since 70% of the data originates from ‘Active' employees, both classification models predicted this label. The F1 scores obtained from Random Forest were 0.70, 0.20, 0.05, and 0.05 for 'Active', 'Resigned', 'Retired', and 'Terminated', respectively. Decision Tree produced a higher F1-Score for ‘Active’ with near zero for all minority classes, thereby revealing more bias towards the majority class.

B. Salary Dominates Importance

The most important feature is 'Salary_INR' with an importance of 95.8% in the Random Forest model and 61.0% in the Decision Tree model. As the same consistency holds across both models, it is clear that all other features indicate noise, but not actual predictors of employee status.

C. Attrition Data Deficit

The model heavily misclassifies ‘Resigned’, ‘Retired’, and ‘Terminated’ employees as ‘Active’, confirming the bias toward the majority class rather than genuine prediction.

VIII. Recommendations

A. Institutional Compensation Standardization

The 57% gap points to something an internal compensation audit could help explain. Wage entropy will be eliminated and labor cost will be optimized by establishing strict, transparent wage bands that are directly related to Experience_Years and functional departments [8], [24].

B. Analytical Feature Enrichment

The HR data architecture has to include behavioral columns, such as manager effectiveness ratings, internal promotion interval, and sentiment scores from localized pulse surveys, in order to increase turnover classification accuracy over 54% [3],[7],[15].

IX. Conclusion

In this project, a large-scale enterprise HR dataset with two million records was subjected to an advanced data analytics and machine learning pipeline. By applying the tree-based classification, both with Random Forest and Decision Tree models, the misalignments in the structure are revealed. Due to its massive entropy, compensation significantly biases both classification models even though it was shown to be highly decentralized, with its variation irrespective of performance or experience. Random Forest outperformed Decision Tree with a higher accuracy for all employee status classes, whereas Decision Tree had a higher overall accuracy of 69%. Future models need to include features that better represent behaviors and class-balancing techniques as a means of creating a complete corporate attrition engine, even though the current models provide operational value in the identification of steady and active employees immediately [15], [16], [19].

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